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Re-evaluation of battery-grade lithium purity toward sustainable batteries

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Recently, the cost of lithium-ion batteries has risen as the price of lithium raw materials has soared and fluctuated. Notably, the highest cost of lithium production comes from the impurity elimination process to satisfy the battery-grade purity of over 99.5%. Consequently, re-evaluating the impact of purity becomes imperative for affordable lithium-ion batteries. In this study, we unveil that a 1% Mg impurity in the lithium precursor proves beneficial for both the lithium production process and the electrochemical performance of resulting cathodes. This is attributed to the increased nucleation seeds and unexpected site-selective doping effects. Moreover, when extended to an industrial scale, low-grade lithium is found to reduce production costs and CO_2 emissions by up to 19.4% and 9.0%, respectively. This work offers valuable insights into the genuine sustainability of lithium-ion batteries.

Lithium-ion batteries (LIBs) have emerged as prevailing energy storage devices for portable electronics and electric vehicles (EVs) because of their exceptionally high-energy density compared with those of other energy storage systems[1]. However, the cost of LIBs, especially that of cathode materials, has been pointed out as the biggest hurdle to achieving affordable EVs. In this context, most cathode studies have been directed toward replacing expensive transition metals (TMs) with low-cost elements while increasing energy densities, for example, the introduction of disordered rocksalt[2,3], Co-free cathodes[4,5], high-Ni layered oxide[6–8], and over-lithiated layered oxide showing anionic redox[9,10]. These strategies have been praised for significantly reducing the cost of cathodes by removing expensive Co elements and enhancing their energy densities. Meanwhile, the lithium price has risen above eightfold from 2020 to the end of 2022 and fluctuated[11] (Fig. 1a) because lithium raw materials exist only in limited countries and due to the sudden changes in international situations[12]. Consequently, the lithium source cost now surpasses that of TM for new classes of cathode materials[13]. For example, as shown in Fig. 1b, the lithium material costs in $LiNi_{0.6}Co_{0.2}Mn_{0.2}O_2$ (NCM622) and $LiNi_{0.8}Co_{0.1}Mn_{0.1}O_2$ (NCM811) used to be 19.6% and 23.5% but are now up to 55.2% and 57.5%, respectively.

The soaring lithium costs naturally motivate us to take a closer look at lithium production, finding a chance to lower LIB prices. Most lithium sources are mined from lithium brines and hard rocks. As illustrated in Fig. 1c, these raw lithium materials are concentrated in liquid phases first. In the case of hard rocks, spodumene of $LiAl(SiO_3)_2$ requires calcination and acid roasting before the concentration process. Then, concentrated solutions are refined to meet the battery-grade purity of over 99.5% (yellowish region) followed by lithium extraction (LX) to produce final products such as Li_2CO_3 (LC) or $LiOH{\cdot}H_2O$ (LH). Mg and Ca are the major impurities for the wet-chemical refinery process, and they are eliminated using the chemical agents $Ca(OH)_2$ and Na_2CO_3, respectively (the details of such processes are described in Supplementary Figs. S1 and S2 for brine and hard rock, respectively). It should be noted that these refinery processes are responsible for 30–40% of the total costs (Supplementary Tables S1 and S2) of lithium production as well as major CO_2 emission (see a yellowish region in Fig. 1c). Moreover, the removal of impurity precipitates in high-purity refining processes is costly due to the presence

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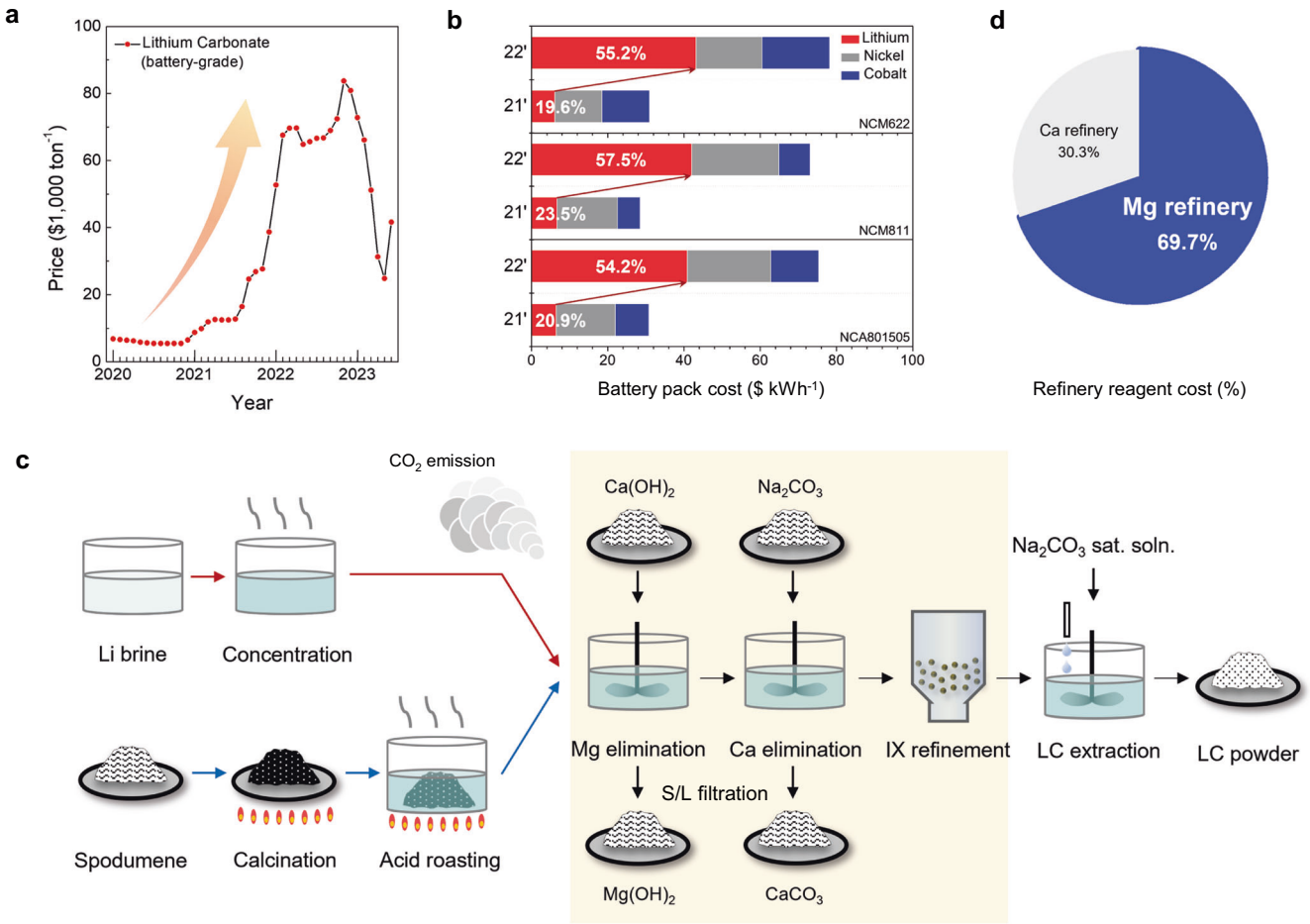

Fig. 1 | Skyrocketing lithium prices and a scheme for lithium extraction processes. a Price history of battery-grade lithium carbonate from 2020 to 2023[11]. **b** Cost breakdown of incumbent cathode materials (NCM622, NCM811, and NCA801505) for lithium, nickel, and cobalt based on material prices in March 2021 and 2022[13]. **c** Simplified process diagram of lithium carbonated production from lithium brine (top) and hard rock (bottom) (IX ion exchange, LC lithium carbonate, LX lithium extraction). **d** Reagent cost ratio of Mg and Ca refineries in lithium extraction processes.

of micron-sized particles. This necessitates expensive facilities, additional processing steps, and filter consumables. Notably, Mg elimination accounts for more than two-thirds of overall refinery costs compared with Ca refinery (Fig. 1d) because of Mg's smaller precipitate particles and high impurity concentrations. Accordingly, refinery works have dedicated themselves to achieving cost-effective and efficient Mg elimination[14–16] from high-Mg/Li-ratio brine[17,18] to meet the battery-grade purity. Furthermore, impurity and quality control in industry-scale cathode material production are gaining increased attention[19].

In this study, we systemically re-evaluated the impacts of impurity grade in terms of lithium extraction processes, electrochemical properties of the resulting cathodes, material production costs, and environmental impacts. We found that Mg impurity of up to 1% in lithium raw materials has unexpected benefits: (i) improvements in flowability and production speed of lithium product through the seeding effect, (ii) cyclability and rate capability enhancements through the anomalous site-selective doping effect, and (iii) significant reductions in expenditures and CO_2 emission ensuing from simpler purification processes. In particular, we would like to emphasize that the solid-solution phase of Li and Mg precursors increased the temperature at which the Mg dopant is incorporated into the oxide structure, leading to an abnormal Li site-selective doping. Consequently, notwithstanding the use of a low-grade lithium source, the electrochemical performances of the resulting cathode were superior to that of conventionally Mg-doped materials. Upon expansion to industry-scale testing, lower-grade lithium sources reduced production costs and CO_2 emissions by 19.4% and 9.0%, respectively, because of the elimination of intensive precipitate removal processes and the decreased purity requirements. These new insights into the lithium grade should drive a paradigm shift toward securing the sustainability of LIBs.

Results

Production of *low-grade* lithium sources

We mimicked the conventional lithium extraction process from brine and hard rock but controlled the Mg^{2+} impurity concentrations systematically to investigate their impact on lithium grade. Here, we produced a carbonate product rather than the hydroxide form as LH requires stringent storage conditions to control air exposure due to the carbonation reaction; in contrast, LC is more widely used as a strategic material in stages that require transport or storage. Bare LC, denoted as LCB, was synthesized by dropping a saturated Na_2CO_3 solution into a Li_2SO_4 solution batch, as shown in Supplementary Note S1. Mg^{2+}-containing LC powders were prepared in the same manner, but the reaction batches contained different amounts of Mg^{2+} ions from 0.1 to 0.9 g L^{-1} (Supplementary Note S1). Inductively coupled plasma optical emission spectrometry (ICP-OES; Supplementary Fig. S4 and Table S3) confirmed the grade of the prepared LC powders, showing a gradual Mg purity variation from 0% to 2.52% depending on the batch Mg concentration. We selected the 0.3 g L^{-1} (0.98% Mg concentration in LC powder) and 0.9 g L^{-1} (2.52% Mg concentration in

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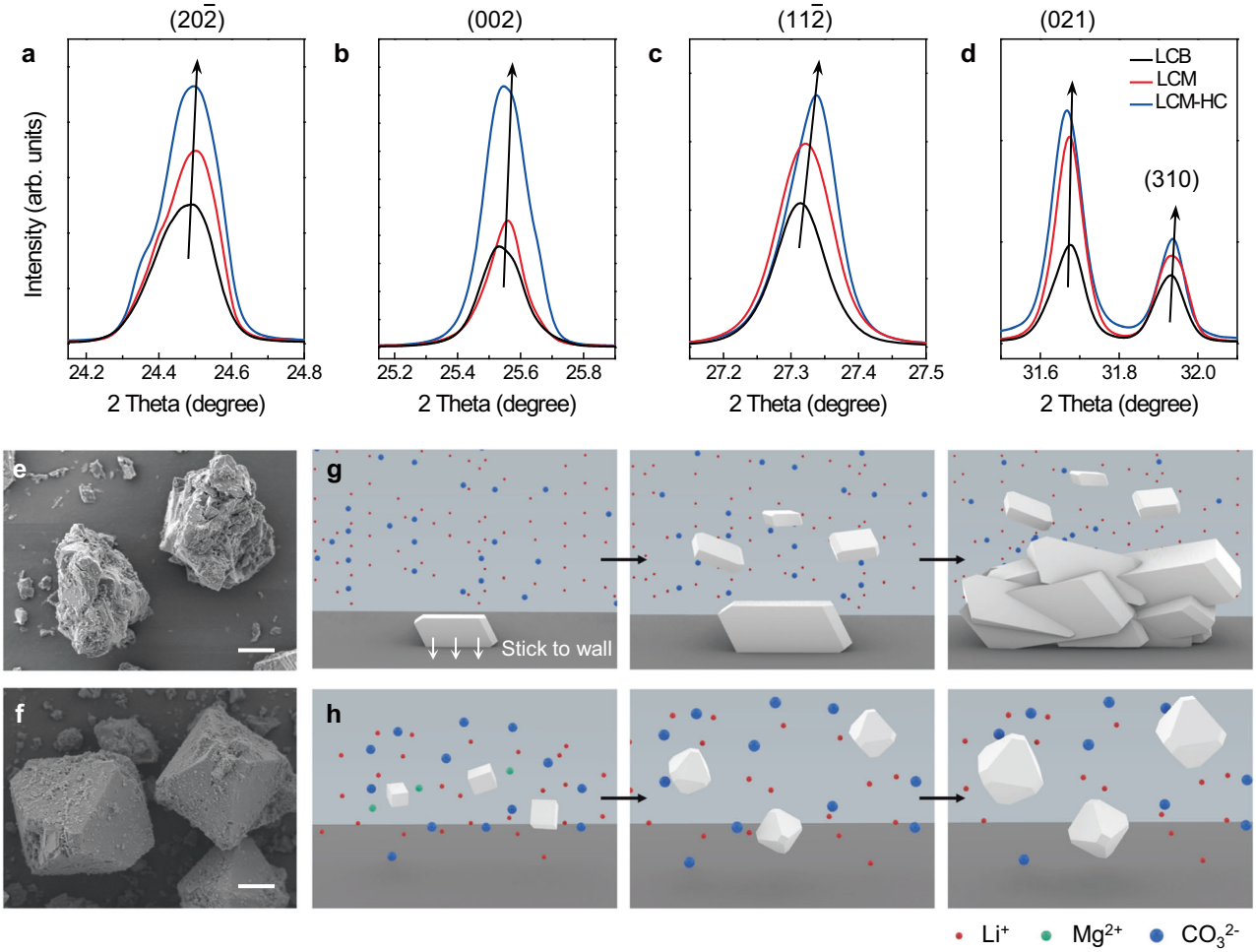

Fig. 2 | Solid-solution behaviour and seeding effect of $MgCO_3$ for Li_2CO_3 synthesis. a–d XRD patterns of the (20$\bar{2}$), (002), (11$\bar{2}$), (021) and (310) planes of LCB, LCM, and LCM-HC powders. High-angle shifts were observed at the (20$\bar{2}$) (**a**), (002) (**b**), (11$\bar{2}$) (**c**), and (310) (**d**) planes with the increasement of peak intensities. **e**, **f** SEM images of LCB (**e**) and LCM-HC (**f**) particles. Scale bars, 10 µm. **g**, **h** Schematic diagrams of the nucleation and growth mechanisms of Li_2CO_3 without (**g**) and with (**h**) Mg^{2+} impurity.

LC powder) batch samples as representatives and denoted them as LCM and LCM-HC, respectively, for better discussion.

In this experiment, we found that the grade control of the lithium extraction process leads to the following distinguishable results: (i) formation of a $MgCO_3$–Li_2CO_3 solid solution, (ii) accelerated lithium extraction reaction rate, and (iii) change in the resulting particle morphology. A comparison of the X-ray diffraction (XRD) patterns of LCB, LCM, and LCM-HC showed a gradual high-angle shift of the Bragg peaks (Fig. 2a–d) with increasing Mg^{2+} concentration. They obeyed Vegard's law well, and a smaller ion radius of Mg^{2+} (0.72 Å) (Li^+ = 0.76 Å) decreased the lattice parameters, suggesting the formation of the $MgCO_3$–Li_2CO_3 solid solution. However, a trace amount of crystalline secondary phase corresponding to $MgCO_3{\cdot}3H_2O$ was observed over the LCM condition, as discussed in Supplementary Note S2. In addition, a preferential increment of the peak intensity related to the *c*-axis was observed (Supplementary Fig. S6) according to impurity concentration, suggesting that the impurity promoted preferential crystal growth.

Mg impurity had a great advantage on the production rate of Li_2CO_3 through its seeding effect (Supplementary Fig. S7). While observing the precipitation process, we noticed that the mixed solution turned cloudy almost immediately after adding a Na_2CO_3-saturated solution to Mg^{2+}-containing batches (Supplementary Fig. S7, black dashed line). In contrast, it took a few minutes for LCB to become cloudy (Supplementary Fig. S7, red dashed line). This supports that the nucleation kinetics of Li_2CO_3 particles is better facilitated with Mg^{2+} ions because of the Mg^{2+} seeding effect as a result of the lower solubility of $MgCO_3$ than that of Li_2CO_3 (the solubilities of Li_2CO_3 and $MgCO_3$ were 6.9×10^{-1} and 6.3×10^{-3} g per 100 ml at 100 °C, respectively). The tracing of the pH change trend diagram well proves this seeding effect of the Mg^{2+} impurity. The diagram shows that the initial pH values were downshifted according to the concentration of Mg^{2+} impurity (Supplementary Fig. S7, red to purple line), indicating that the number of seeds (nucleation sites of Li_2CO_3) increased as the Mg^{2+} concentration increased, thus resulting in early-stage proton generation (see chemical reaction (5)).

The inclusion of impurity also led to well-defined particle morphologies (Fig. 2e, f) and narrowed particle size distributions (PSD) verified using scanning electron microscopy (SEM; Supplementary Fig. S8) and ex-situ PSD analysis (Supplementary Fig. S9). Generally, battery-grade Li_2CO_3 (Fig. 2g) slowly nucleated, particularly on the surface of the reaction batch and impeller, because of its lack of nucleation mediation. Primary particles tended to agglomerate and grow, making a secondary particle morphology with a needle-like primary particle shape (Fig. 2e) with inhomogeneous PSD[20]. Meanwhile, the introduction of Mg impurity resulted in an octahedron morphology (Fig. 2f and Supplementary Figs. S10 and S11), which supports the preferred crystal growth confirmed by the XRD patterns (Supplementary Fig. S6). For Li_2CO_3 synthesized with Mg^{2+} impurity (Fig. 2h), $MgCO_3{\cdot}xH_2O$ nanoparticles first precipitated because of its